

Today's Agenda:-

1. Intro To Number System
2. Binary To Decimal & Decimal To Binary
3. Addition of Binary Numbers.
4. Bitwise Operations.
5. Negative Numbers
6. Ranges (int & long).
7. Importance of Constraints.

Decimal Number System

- ↓
10
- Base 10 (Digits 0 to 9)
 - One we use in everyday lives.
 - Every digit represents a certain power of 10.

Ex:-

$$342 = 300 + 40 + 2 = 3 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$$

$$\begin{aligned} 2563 &= 2000 + 500 + 60 + 3 \\ &= 2 \times 10^3 + 5 \times 10^2 + 6 \times 10^1 + 3 \times 10^0 \end{aligned}$$

Binary Number System

↓
2

→ Base 2

→ Only 2 digits (0 & 1)

→ Used in electronic systems

→ Every digit can be represented using a power of 2.

Ex:-

$$\begin{array}{cccc} 2 & 1 & 0 & \\ 1 & 1 & 0 & \end{array} = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$
$$= 4 + 2 + 0 = 6.$$

$$\begin{array}{cccc} 3 & 2 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{array} = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$
$$= 8 + 0 + 2 + 1$$
$$= \underline{\underline{11.}}$$

Binary To Decimal Conversion

Ex:-

3	2	1	0
1	1	0	1

			$\rightarrow 1 \times 2^0 = 1$
		$\rightarrow 0 \times 2^1 = 0$	
	$\rightarrow 1 \times 2^2 = 4$		
$\rightarrow 1 \times 2^3 = 8$			
			<u>13</u>

Ex:-

4	3	2	1	0
1	0	1	0	1

				$\rightarrow 1 \times 2^0 = 1$
			$\rightarrow 0 \times 2^1 = 0$	
		$\rightarrow 1 \times 2^2 = 4$		
	$\rightarrow 0 \times 2^3 = 0$			
$\rightarrow 1 \times 2^4 = 16$				
				<u>21</u>

Quiz 1

6	5	4	3	2	1	0
1	0	1	1	0	1	0

						$\rightarrow 0 \times 2^0 = 0$
					$\rightarrow 1 \times 2^1 = 2$	
			$\rightarrow 0 \times 2^2 = 0$			
		$\rightarrow 1 \times 2^3 = 8$				
	$\rightarrow 1 \times 2^4 = 16$					
$\rightarrow 0 \times 2^5 = 0$						
$\rightarrow 1 \times 2^6 = 64$						
						<u>90</u>

Decimal To Binary Conversion

→ Using long division method

Ex :- $(20)_{10} = (?)_2$

2	20	0
2	10	0
2	5	1
2	2	0
2	1	1
	0	

$(10100)_2$

Quiz 2

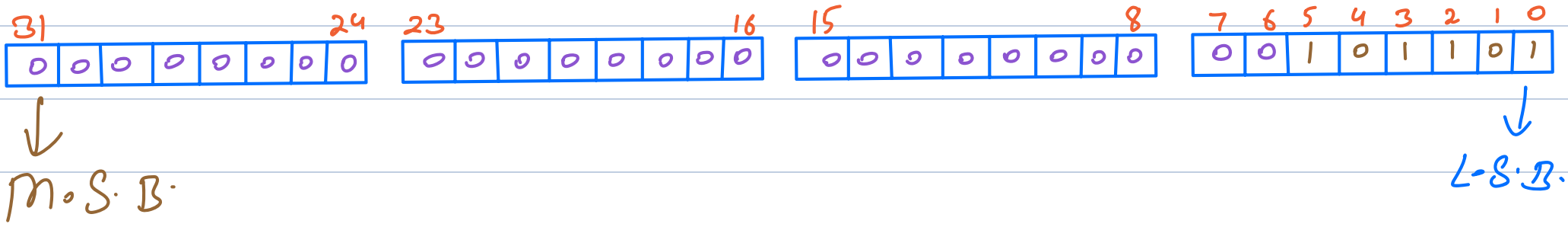
$(45)_{10} = (?)_2$

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
2	1	1
	0	

$(101101)_2$

4 B; 1 B = 8 bits
32 bits

int x = 45



MSB = 0 → No. is positive

1 → No. is negative.

Addition of Decimal Numbers

$$\begin{array}{r}
 \begin{array}{ccc}
 +1 & +1 & \\
 3 & 6 & 8 \\
 + & 2 & 5 & 3 \\
 \hline
 6 & 2 & 1
 \end{array}
 \end{array}$$

digit	Carry
11 · 1 · 10	11 / 10
12 · 1 · 10	12 / 10
6 · 1 · 10	6 / 10

Addition of Binary Numbers

Ex:-

$$\begin{array}{r}
 \begin{array}{cccccc}
 +1 & +1 & +1 & +0 & +1 & \\
 1 & 0 & 1 & 0 & 1 & \\
 + & 1 & 1 & 0 & 1 & \\
 \hline
 1 & 0 & 0 & 0 & 1 & 0
 \end{array}
 \end{array}$$

digit	Carry
2 · 1 · 2	2 / 2
1 · 1 · 2	1 / 2
2 · 1 · 2	2 / 2
2 · 1 · 2	2 / 2
2 · 1 · 2	2 / 2
1 · 1 · 2	1 / 2

Q:-3

$$\begin{array}{r}
 \begin{array}{cccccc}
 0 & 0 & 1 & 1 & 0 & \\
 1 & 0 & 1 & 1 & 0 & \\
 + & 0 & 0 & 1 & 1 & 1 \\
 \hline
 1 & 1 & 1 & 0 & 1 &
 \end{array}
 \end{array}$$

digit	Carry
1 · 1 · 2	1 / 2
2 · 1 · 2	2 / 2
3 · 1 · 2	3 / 2
1 · 1 · 2	1 / 2
1 · 1 · 2	1 / 2

Bitwise Operators ($&$, $|$, $^$, $\sim$) $\gg$, $\ll$

→ Operators used to perform operations on individual bits of binary numbers.

→ Used to manipulate binary data

→ $0 \rightarrow$ false / unset
 $1 \rightarrow$ true / set

AND ($\&$)

→ Performs Logical AND

→ Return true if both bits are set, else false.

↓
1

			<u>Result</u>
0	&	0	0
0	&	1	0
1	&	0	0
1	&	1	1

OR (|)

- Performs Logical OR
- Return true if either of bits are set, else false.

		<u>Result</u>
0	0	0
0	1	1
1	0	1
1	1	1

XOR (^)

- Perform exclusive OR.
- Return true if both bits are different, else false.

		<u>Result</u>
0	0	0
0	1	1
1	0	1
1	1	0

NOT (~) !

- Operates on a single bit
- Toggles it.

	<u>Result</u>
~ 0	1
~ 1	0

Examples

①

$$5 \times 6$$

$$= \underline{\underline{4}}$$

$$\begin{array}{r} \leftarrow \\ 101 \\ \times 110 \\ \hline 100 \end{array}$$

②

$$20 \times 45$$

$$= \underline{\underline{4}}$$

$$\begin{array}{r} 010100 \\ \times 101101 \\ \hline 000100 \end{array}$$

③ $92 \wedge 154$

$= 198$

$$\begin{array}{r}
 \begin{array}{cccccccc}
 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\
 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0
 \end{array} \\
 \hline
 \begin{array}{cccccccc}
 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0
 \end{array} \\
 \hline
 \begin{array}{cccccccc}
 7 & 6 & & & & 2 & 1 & \\
 128 & 64 & & & & 4 & 2 &
 \end{array}
 \end{array}$$

④ ~ 01011100

$= 0100011$

⑤ $92 \mid 154$

$= 222$

$$\begin{array}{r}
 \begin{array}{cccccccc}
 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\
 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0
 \end{array} \\
 \hline
 \begin{array}{cccccccc}
 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0
 \end{array} \\
 \hline
 \begin{array}{cccccccc}
 7 & 6 & & 4 & 3 & 2 & 1 & \\
 128 & 64 & & 16 & 8 & 4 & 2 &
 \end{array}
 \end{array}$$

Quiz 4

$20 \wedge 45$

$= 57$

$$\begin{array}{r}
 \begin{array}{cccccc}
 0 & 1 & 0 & 1 & 0 & 0 \\
 1 & 0 & 1 & 1 & 0 & 1
 \end{array} \\
 \hline
 \begin{array}{cccccc}
 1 & 1 & 1 & 0 & 0 & 1
 \end{array} \\
 \hline
 \begin{array}{cccccc}
 5 & 4 & 3 & & & 0 \\
 32 & 16 & 8 & & & 1
 \end{array}
 \end{array}$$

Binary Representation of Negative Numbers

ex1- Represent (-5) in 8 bit signed format.

Three Step Process: (Taking 2's complement)

1. Convert the abs. value of No. to binary.

7	6	5	4	3	2	1	0
0	0	0	0	0	1	0	1

2. Reverse all the bits.

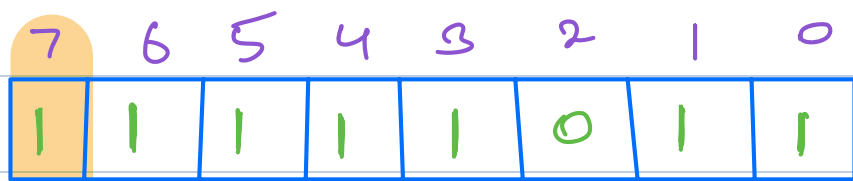
7	6	5	4	3	2	1	0
1	1	1	1	1	0	1	0

3. Add 1 to the result of step 2.

7	6	5	4	3	2	1	0
1	1	1	1	1	0	1	0
+ 0 0 0 0 0 0 0 1							

1 1 1 1 1 0 1 1

→ $(-5)_{10}$



1×2^0	=	1
1×2^1	=	2
0×2^2	=	0
1×2^3	=	8
1×2^4	=	16
1×2^5	=	32
1×2^6	=	64
$-(1 \times 2^7)$	=	-128
		-5

Quiz 5

-3 in 8-bit signed integer format.

1.

7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	1

2.

7	6	5	4	3	2	1	0
1	1	1	1	1	1	0	0

+1

3.

7	6	5	4	3	2	1	0
1	1	1	1	1	1	0	1

Quiz 6

-10 in 8-bit signed integer format.

1.

7	6	5	4	3	2	1	0
0	0	0	0	1	0	1	0

2.

7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	1

3.

7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	1

+1

1	1	1	1	0	1	1	0
---	---	---	---	---	---	---	---

8:20

Range of Data Types (Signed)

3	2	1	0
1	0	0	0

$$= -2^3$$

3	2	1	0
0	1	1	1

$$\begin{aligned} &2^0 + 2^1 + 2^2 \\ &= 2^0 \left(\frac{2^3 - 1}{2 - 1} \right) \\ &= 2^3 - 1 \end{aligned}$$

$$\frac{-8 \text{ to } +7}{-2^3 \text{ to } 2^3 - 1}$$

7	6	5	4	3	2	1	0
1	0	0	0	0	0	0	0

$$= -2^7$$

7	6	5	4	3	2	1	0
0	1	1	1	1	1	1	1

$$\begin{aligned} &2^0 + 2^1 + \dots + 2^6 \\ &= 2^7 - 1 \end{aligned}$$

$$\frac{-2^7 \text{ to } 2^7 - 1}{-128 \text{ to } 127.}$$

Generalising for a N bit data type:

$$\text{Range} = [-2^{N-1}, +2^{N-1} - 1].$$

Now,

For an int data type; (4B)
 $N = 32;$

$$\therefore \text{Range of } \text{int} = [-2^{31}, 2^{31} - 1]$$

$$\approx [-2 \times 10^9, 2 \times 10^9]$$

Similarly for long, $N = 64$:

$$\text{Range of long} = [-2^{63}, 2^{63} - 1]$$

$$\approx [-9 \times 10^{18}, 9 \times 10^{18}]$$

Importance of Constraints

int a = 10^5
int b = 10^6

Try to compute $a * b$;

1. int c = $a * b$; X;

2. long c = $a * b$; X;

ALU

3. long c = long(a) * b; ✓

long c = (long) a * b; ✓
long ml int

Q:- $1 \leq N \leq 10^5$
 $1 \leq A[i] \leq 10^6$

Given an array of size N , Cal. the sum of array elements.

```
long.  
int sum = 0;  
for (i = 0; i < N; i++) {  
    sum += arr[i];  
}  
return sum;
```

Contest $\rightarrow$ 14th May
7:00 AM IST.

90 mins.

4 Questions.

1 Assignment
1 Easy
1 Easy - Med.
1 Med Hard.

3/4 Questions.

$R_1 \rightarrow T+1 \rightarrow T+8$
 $R_2 \rightarrow T+9 \rightarrow T+15$
 $R_3 \rightarrow T+16 \rightarrow T+22$

$$\begin{array}{|c|} \hline 1011011 \\ \hline \end{array}$$

$$\begin{array}{r} 1010011 \\ 1001001 \\ \hline 81000001 \end{array}$$

$$10011010$$

$$a \times 1 = 1.$$

$$1 \times 2^0 + \dots$$